# Chapter 9 — CSS Grid

Flexbox (Chapter 8) arranges items along **one** axis and lets their content decide the sizes. **CSS Grid** is the other half of modern layout: you define **rows and columns at the same time**, then place items into the cells — or let the browser place them for you. It is the right tool for page structure, card galleries, dashboards (screens of charts and key numbers), forms with aligned labels, and any design that a designer drew on a grid. Grid has worked in every major browser since 2017 (Baseline widely available — Baseline is the label for features that work in all major browsers); `subgrid`, added later, is Baseline since September 2023.

**What you'll learn**

| # | Topic | You will be able to… |
|---|---|---|
| 9.1 | Grid vocabulary | name lines, tracks, cells, areas and gaps, and turn a container into a grid |
| 9.2 | Defining tracks | size tracks with px, %, `fr`, `auto`, `min-content`, `max-content`, `fit-content()`, `minmax()` — and do the `fr` maths by hand |
| 9.3 | `repeat()`, `auto-fill` vs `auto-fit` | build a responsive grid (one that adapts to the screen width) with no media queries (CSS rules that apply only at certain screen sizes, Chapter 10) and predict its column count |
| 9.4 | Placing items | use line numbers, `span`, negative lines, named lines, `grid-template-areas` and overlap |
| 9.5 | The implicit grid and auto-placement | control extra rows/columns, `grid-auto-flow` and `dense` |
| 9.6 | Alignment | align items in cells and the whole grid in its container |
| 9.7 | Subgrid | line up content across sibling cards |
| 9.8 | Grid vs Flexbox | choose the right tool (and combine them) |
| 9.9 | Patterns | holy grail layout (the classic header + three columns + footer page), responsive card grid, overlays (one element laid over another) |
| 9.10 | Debugging, pitfalls and interview prep | find and fix the common grid bugs |

As in Chapters 7 and 8, every example is a page in `code/` with a screenshot in `img/`: grid containers have a **dashed purple border**, items a **solid blue outline**, **red badges** mark numbered grid lines and **red numbers** are track sizes the page read back from the browser. The `.mjs` scripts read `getComputedStyle(grid).gridTemplateColumns` (`getComputedStyle` is the JavaScript function that returns an element's final styles), which returns the **resolved** track sizes (the final numbers after the browser has done the maths) in pixels — the browser's own answer — and compare it with the hand calculation.

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## 9.1 Grid vocabulary

### What it is

`display: grid` (or `inline-grid`) makes an element a **grid container**; its direct children become **grid items**. The container defines an invisible grid:

| Term | Meaning |
| --- | --- |
| **Grid line** | A dividing line between tracks. Numbered from **1** at the start edge; also numbered backwards from **−1** at the end edge. A grid with 3 columns has 4 column lines. |
| **Track** | The space between two adjacent lines: a **column** or a **row**. |
| **Cell** | The intersection of one row and one column — the smallest unit. |
| **Grid area** | Any rectangle of cells bounded by four lines (can be named). |
| **Gutter / gap** | Space between tracks, set with `gap`, `row-gap`, `column-gap`. Gaps exist only *between* tracks, never at the edges. |
| **Explicit grid** | The tracks you declared with `grid-template-*`. |
| **Implicit grid** | Extra tracks the browser creates when items are placed outside the explicit grid, or when there are more items than cells. |

Analogy: a spreadsheet. The column and row dividers are the lines, a column is a track, a cell is a cell, and a merged range of cells is a grid area. Unlike a spreadsheet, you size the columns with maths (`fr`, `minmax`) and one item may span several cells.

### Grid behaviours to know up front

- As with flex items, grid items are blockified (treated as block boxes, so even a `<span>` accepts width and height), `float` is ignored, margins don't collapse, and `z-index` works without `position` (Chapter 7).
- Grid is **layout-in**: tracks are defined by the container and items fall into them. Flexbox is **content-out**: the items' sizes drive the layout. That difference drives the choice in 9.8.

#### Self-check

1. How many column lines does `grid-template-columns: 1fr 1fr 1fr 1fr` create? → 5.
2. Which line is `-1`? → The last line (the end edge of the explicit grid).

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## 9.2 Defining tracks

#### What it is

`grid-template-columns` and `grid-template-rows` list the size of each track, separated by spaces. Values you can use per track:

| Value | Meaning |
| --- | --- |
| `200px`, `10rem`, `30%` | Fixed size (percent of the container's size) |
| `fr` | A **fraction of the leftover space** after fixed tracks and gaps are taken out |
| `auto` | As a maximum: content size, but stretches to fill leftover space when there are no `fr` tracks (with `align/justify-content: normal`). As a minimum: the item's automatic minimum size (usually its min-content size, so content is not crushed) |
| `min-content` | As narrow as the content can get (longest word) |
| `max-content` | As wide as the content wants, no wrapping |
| `fit-content(150px)` | `max-content`, but capped at 150px (and at least `min-content`) |
| `minmax(min, max)` | Any size between a min and a max, e.g. `minmax(200px, 1fr)` |

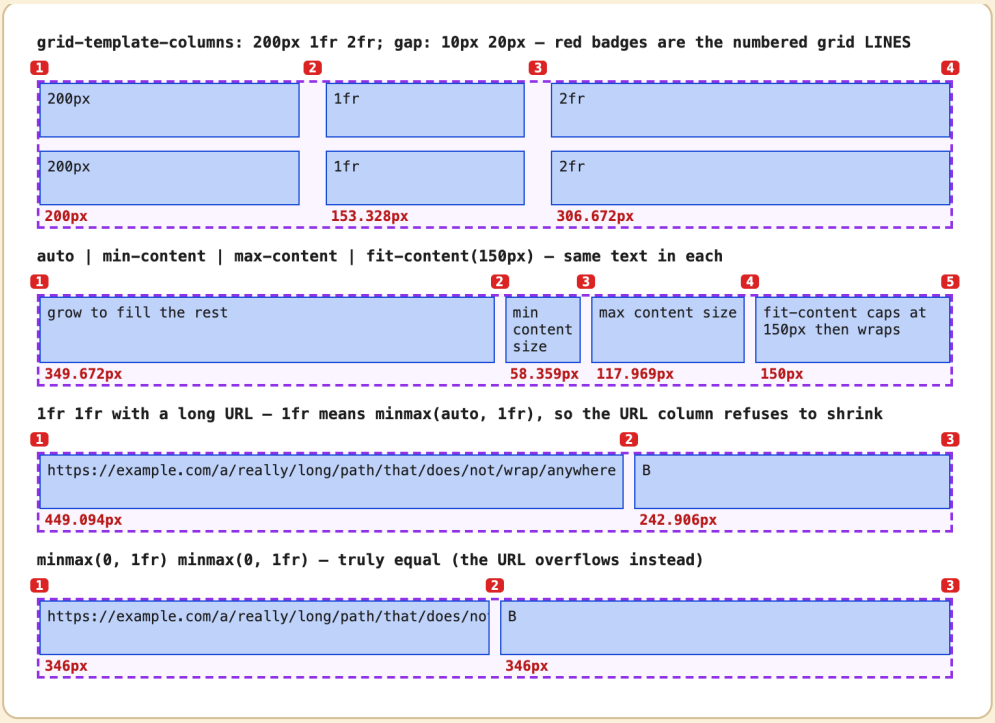



*Figure 9.1 — `code/01_tracks.html`. A script in the page reads `getComputedStyle(grid).gridTemplateColumns` and draws the line numbers and sizes; they are absolutely positioned, so they don't change the layout.*

### How `fr` is calculated

1. Start with the container's inner size (700px here).
2. Subtract every **fixed** track and every gap. For `200px 1fr 2fr` with `column-gap: 20px`: 700 − 200 − 2 × 20 = **460px** of free space.
3. Add up the `fr` values: 1 + 2 = 3, so **1fr = 460 / 3 = 153.33px**, 2fr = 306.67px.
4. (A track whose share would be smaller than its minimum — see below — is fixed at that minimum and the rest is re-split, like flexbox's freeze loop in §8.3.)

`code/01_tracks.mjs` prints the declared value next to what Chrome resolved, then the hand calculation:

```
const rule = [...document.styleSheets[0].cssRules].find(r => r.selectorText === '#' + g.id);
return `${g.id}  ${rule.style.gridTemplateColumns} -> ${getComputedStyle(g).gridTemplateColumns}`;
```

Output:

```
container content width: 700px
t1  200px 1fr 2fr                                   -> 200px 153.328px 306.672px
t2  auto min-content max-content fit-content(150px)  -> 349.672px 58.3594px 117.969px 150px
t3  1fr 1fr                                         -> 449.094px 242.906px
t4  minmax(0px, 1fr) minmax(0px, 1fr)               -> 346px 346px

by hand, t1: 1fr = 460 / 3 = 153.333px, 2fr = 306.667px (Chrome stores 1/64 px: 153.328125, 306.671875)
```

The tiny differences are Chrome's 1/64-pixel layout units, exactly as in the flexbox chapter.

### `fr` vs `%`

`%` tracks are a share of the *whole* container and ignore gaps, so `50% 50%` plus a gap overflows. `fr` shares only what's left after gaps: `1fr 1fr` never overflows because of gaps.

### The `1fr` minimum trap (t3 vs t4)

`1fr` is shorthand for **`minmax(auto, 1fr)`**. The `auto` minimum means "never narrower than the item's min-content size". So in t3 a long unbreakable URL forces its column to 449px and the other gets the remaining 242.9px — the columns are not equal. It is the grid twin of flexbox's `min-width: auto` problem (§8.4).

Fix: `minmax(0, 1fr)` (t4) lets the track shrink below its content — the columns become exactly 346px each, and you handle the overflowing content with `overflow`, `overflow-wrap: anywhere` or `min-width: 0` on the item. Rule of thumb: for columns that **must** be equal (a dashboard, a code-and-preview split), write `repeat(3, minmax(0, 1fr))`.

### Common mistakes

- Using `%` tracks together with `gap` and getting a horizontal scrollbar (the page becomes wider than the screen and scrolls sideways).
- Expecting `1fr 1fr` to be equal whatever the content (see above).
- Forgetting that rows default to `auto` — a row only gets taller than its content if you give it a size or the container a height with `fr` rows.

### Self-check

1. Container 1000px, `grid-template-columns: 250px 1fr 1fr 1fr; column-gap: 10px`. Size of 1fr? → (1000 − 250 − 30) / 3 = **240px**.
2. What is `1fr` short for? → `minmax(auto, 1fr)`.
3. Which track value never grows beyond its content or a cap? → `fit-content(cap)`.

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## 9.3 `repeat()`, `auto-fill` and `auto-fit`

### What it is

`repeat(count, tracks)` saves typing: `repeat(4, 1fr)` = `1fr 1fr 1fr 1fr`; `repeat(2, 100px 1fr)` = `100px 1fr 100px 1fr`.

Instead of a number, the count can be `auto-fill` or `auto-fit`: "as many tracks as fit". Combined with `minmax()` this gives a responsive grid with **no media queries** — often called the RAM pattern (**R**epeat, **A**uto, **M**inmax):

```
.gallery { display: grid; gap: 10px;
           grid-template-columns: repeat(auto-fill, minmax(120px, 1fr)); }
```

### How it works

1. The browser computes how many tracks of the *minimum* size fit: **n = floor((width + gap) / (min + gap))**.
2. It creates n tracks, then the `1fr` maximum stretches them to share the leftover space.
3. Items are auto-placed into them; extra items wrap onto new rows.
4. The difference appears only when there are **fewer items than tracks**:
   - `auto-fill` keeps the empty tracks — items stay at their "one-column" width and space is reserved on the right.
   - `auto-fit` **collapses** empty tracks to 0 (and the gaps next to them), so the items stretch to fill the row.

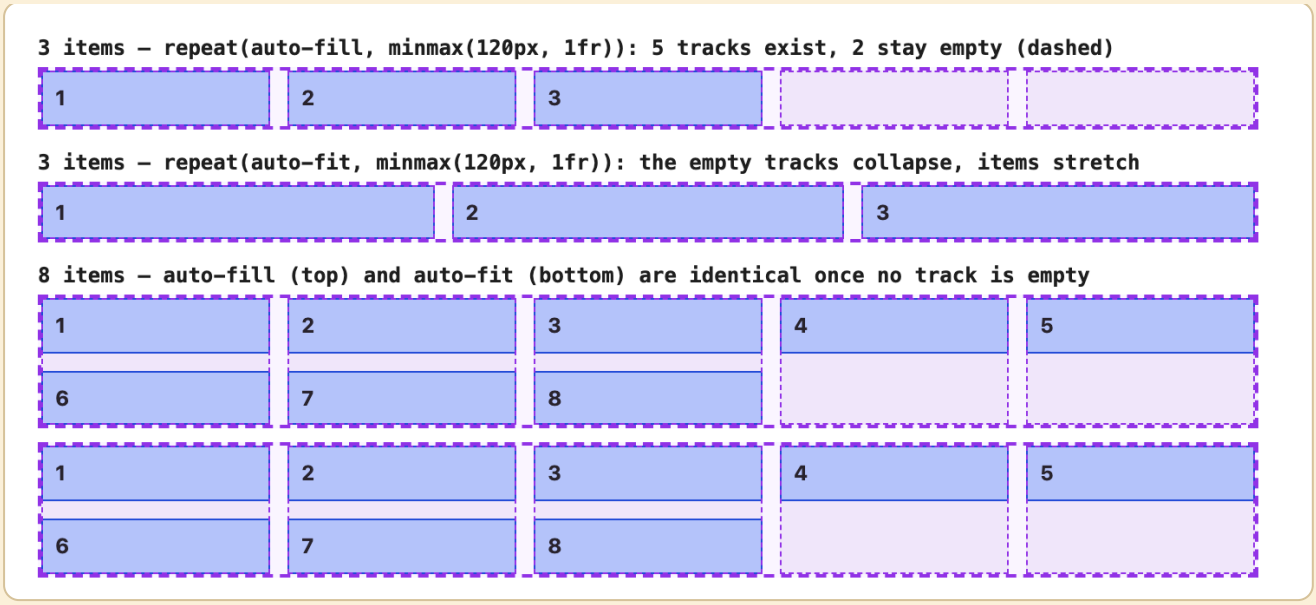



*Figure 9.2 — `code/02_auto_fill_fit.html`. Every track, including empty ones, is outlined in dashed purple from the resolved track list.*

Output of `code/02_auto_fill_fit.mjs`:

```
container 700px, gap 10px, minmax(120px, 1fr):
  fill3  132px 132px 132px 132px 132px
  fit3   226.656px 226.672px 226.672px 0px 0px
  fill8  132px 132px 132px 132px 132px
  fit8   132px 132px 132px 132px 132px
container 400px, gap 10px, minmax(120px, 1fr):
  fill3  126.656px 126.672px 126.656px
  fit3   126.656px 126.672px 126.656px
  fill8  126.656px 126.672px 126.656px
  fit8   126.656px 126.672px 126.656px
by hand, 700px: n = floor(710 / 130) = 5, track = (700 - 40) / 5 = 132.000px
by hand, 400px: n = floor(410 / 130) = 3, track = (400 - 20) / 3 = 126.667px
by hand, auto-fit with 3 items at 700px: empty tracks and their gaps collapse -> (700 - 20) / 3 = 226.667px
```

Note `fit3`: the two collapsed tracks show up as `0px` in the resolved list.

### Which one to use

| Want | Use |
|---|---|
| Cards keep a consistent size even when there are only one or two | `auto-fill` |
| A few items should stretch to fill the row | `auto-fit` |

### Guarding the minimum on narrow screens

If the container gets narrower than the minimum (a 120px track on a 100px-wide phone column), the single track overflows. Write the minimum as `min(120px, 100%)` so it can never exceed the container: `repeat(auto-fit, minmax(min(160px, 100%), 1fr))` — used in 9.9.

### Self-check

1. Container 1000px, gap 20px, `repeat(auto-fill, minmax(200px, 1fr))`. How many columns and how wide? → floor(1020 / 220) = 4 columns of (1000 − 60) / 4 = **235px**.
2. Same, 2 items, `auto-fit`. Width of each? → Two tracks remain, (1000 − 20) / 2 = **490px**.
3. Can `auto-fill` be combined with a plain `1fr` (`repeat(auto-fill, 1fr)`)? → No — auto-repeat needs a definite size (one known before layout, not dependent on leftover space) (a fixed length or a `minmax` with a fixed minimum) to know how many fit.

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## 9.4 Placing items

By default each item goes into the next free cell (9.5). To put an item somewhere specific you name its **start and end lines** on each axis.

### Line-based placement

| Property | Example | Meaning |
|---|---|---|
| `grid-column-start` / `grid-column-end` | `1` / `3` | Start at column line 1, end at line 3 (2 columns wide) |
| `grid-column` (shorthand) | `1 / 3` | Same |
| `span n` | `grid-column: span 2` | 2 tracks wide, starting wherever auto-placement puts it |
| Negative lines | `grid-column: 1 / -1` | From the first to the **last** line: full width |
| `grid-row` | `1 / span 2` | Start at row line 1, 2 rows tall |
| `grid-area` (4 values) | `3 / 2 / 4 / 5` | **row-start / column-start / row-end / column-end** (anticlockwise from the top, unlike `margin`) |

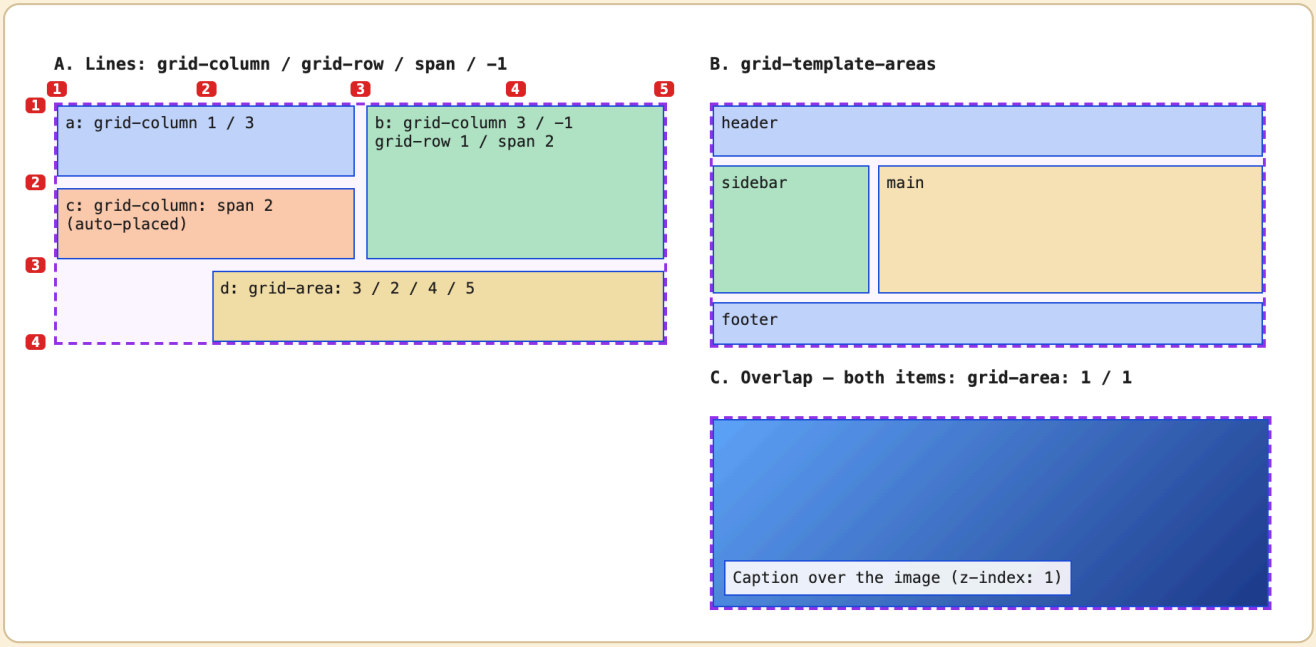



*Figure 9.3 — `code/03_placement.html`.*

```
#lines    { grid-template-columns: repeat(4, 1fr); grid-template-rows: repeat(3, 50px); gap: 8px; }
#lines .a { grid-column: 1 / 3; }
#lines .b { grid-column: 3 / -1; grid-row: 1 / span 2; }
#lines .c { grid-column: span 2; }            /* only a span: auto-placed into the first free spot */
#lines .d { grid-area: 3 / 2 / 4 / 5; }
```

`code/03_placement.mjs` maps each item's rectangle back to line numbers:

```
a: columns 1 -> 3, rows 1 -> 2
b: columns 3 -> 5, rows 1 -> 3
c: columns 1 -> 3, rows 2 -> 3
d: columns 2 -> 5, rows 3 -> 4
```

`-1` resolved to line 5 (the last line of a 4-column grid). Note: negative numbers count from the end of the **explicit** grid only, so `1 / -1` doesn't reach into implicit columns.

### Named lines

You can name lines in square brackets and place by name — useful in big layouts where numbers change:

```
.layout { grid-template-columns: [full-start] 1fr [content-start] minmax(0, 60rem) [content-end] 1fr [full-end]; }
.layout > *      { grid-column: content; }   /* "content" = content-start / content-end */
.layout > .bleed { grid-column: full; }      /* a full-width image or banner */
```

This "full-bleed" layout keeps text in a readable centre column while chosen elements run edge to edge.

`grid-template-areas`

The most readable way to build page layouts: draw the layout as ASCII art, one string per row, then assign items to areas by name.

```
#areas { grid-template-columns: 110px 1fr;
         grid-template-rows: 36px 90px 30px;
         grid-template-areas:
           "header  header"
           "sidebar main"
           "footer  footer"; }
.h { grid-area: header; } .s { grid-area: sidebar; } .m { grid-area: main; } .f { grid-area: footer; }
```

Rules:

- Every string must have the same number of cells.
- A name repeated in adjacent cells makes one area; each area must be a **rectangle** (no L-shapes).
- A `.` (or several dots, `...`) marks an empty cell.
- Named areas automatically create named lines `header-start`/`header-end`, so `grid-column: header` also works.
- Rearranging the layout at a breakpoint (a screen width at which a media query switches the layout) is just rewriting the strings (9.9).

### Overlap

Two items may occupy the same cells. Later items paint on top; `z-index` changes that (it works on grid items without `position`). Demo C puts an image and a caption in `grid-area: 1 / 1` and aligns the caption to the bottom-left with `align-self: end; justify-self: start` — a hero banner (the large image-and-headline block at the top of a page) with no `position: absolute`, so the container still grows with its content.

### Accessibility note

As with flexbox `order`, visual placement doesn't change DOM order (the order of elements in the HTML): keyboard and screen-reader (software that reads the page aloud) users follow the source (§8.2). Keep source order logical; don't use placement to reorder meaningful content.

#### Self-check

1. Make an item span the full width regardless of column count. → `grid-column: 1 / -1`.
2. What does `grid-area: 2 / 1 / 4 / 3` mean? → Rows 2 to 4 (2 rows), columns 1 to 3 (2 columns).
3. Is this area template valid: `"a a" "a b"`? → No — area `a` would be L-shaped.

---

## 9.5 The implicit grid and auto-placement

#### What it is

When there are more items than explicit cells, or an item is placed at a line beyond the explicit grid, the browser adds **implicit tracks**. Their sizes come from `grid-auto-rows` and `grid-auto-columns` (default `auto`).

**Auto-placement** fills cells in order. `grid-auto-flow` sets how:

| Value | Behaviour |
|---|---|
| `row` (default) | Fill each row left to right, then add rows |
| `column` | Fill each column top to bottom, then add columns |
| `row dense` / `column dense` | Also go **back** to fill earlier holes with later items that fit |

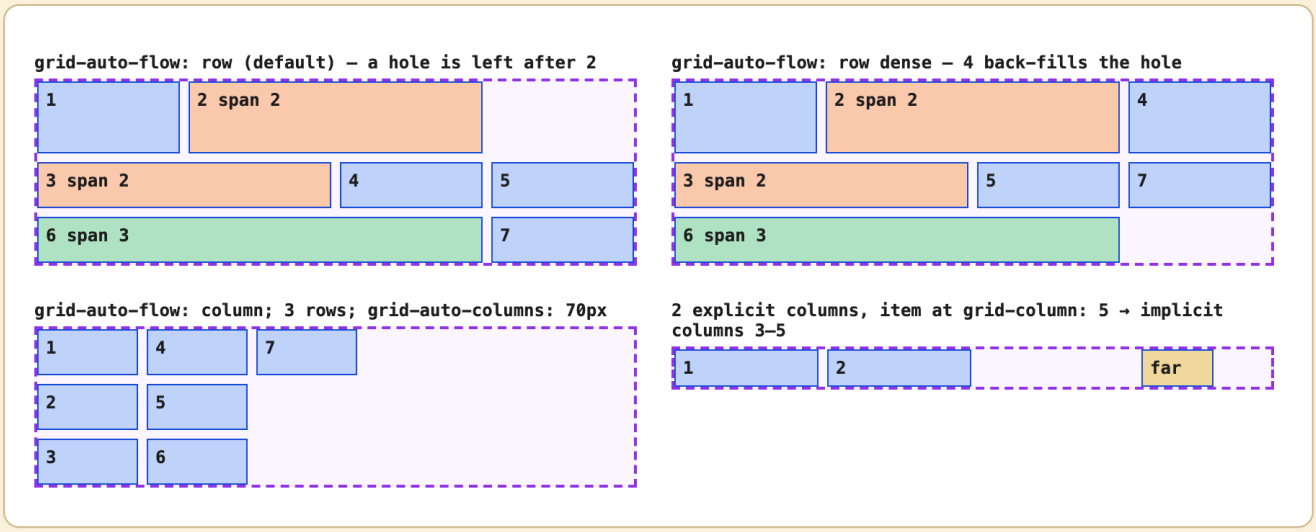



*Figure 9.4 — `code/04_implicit.html`. Orange items span 2 columns, green spans 3.*

```
.flow   { grid-template-columns: repeat(4, 1fr); grid-template-rows: 50px; grid-auto-rows: 32px; }
.dense  { grid-auto-flow: row dense; }
.by-col { grid-template-rows: repeat(3, 32px); grid-auto-flow: column; grid-auto-columns: 70px; }
.outside      { grid-template-columns: repeat(2, 100px); grid-auto-columns: 50px; }
.outside .far { grid-column: 5; }
```

Output of `code/04_implicit.mjs` (`r2c3` = row 2, column 3):

```
row      columns: 99px 99px 99px 99px                      rows: 50px 32px 32px
         placement: 1@r1c1 2@r1c2 3@r2c1 4@r2c3 5@r2c4 6@r3c1 7@r3c4
dense    columns: 99px 99px 99px 99px                      rows: 50px 32px 32px
         placement: 1@r1c1 2@r1c2 3@r2c1 4@r1c4 5@r2c3 6@r3c1 7@r2c4
bycol    columns: 70px 70px 70px                           rows: 32px 32px 32px
         placement: 1@r1c1 2@r2c1 3@r3c1 4@r1c2 5@r2c2 6@r3c2 7@r1c3
outside  columns: 100px 100px 50px 50px 50px               rows: 26px
         placement: 1@r1c1 2@r1c2 far@r1c5
```

Read it like this:

- **`row`**: item 3 needs two columns; only column 4 is free in row 1, so it moves to row 2, leaving **`r1c4` empty**. Only row 1 is explicit (50px); rows 2–3 are implicit (32px from `grid-auto-rows`).
- **`dense`**: item 4 (1 column) goes back and fills r1c4. The items are now drawn **out of source order**.
- **`bycol`**: three explicit rows, columns created as needed at 70px.
- **`outside`**: placing an item at column line 5 creates implicit columns 3, 4 and 5 at 50px each.

### Useful settings

- `grid-auto-rows: minmax(100px, auto)` — rows at least 100px, but grow with content. The standard for card grids.
- `dense` is fine for image galleries where order doesn't matter; avoid it where reading or tab order does — the visual order no longer matches the DOM.

## Self-check

1. `grid-template-columns: repeat(3, 1fr)` and 7 items, no rows declared. How many rows, and how are they sized? → 3 implicit rows, sized by `grid-auto-rows` (default `auto` = content).
2. What does `dense` risk? → Visual order differing from DOM/focus order.

## 9.6 Alignment

Grid uses the same Box Alignment properties (the shared CSS alignment module, §8.5) as flexbox, but here **both axes have a `justify-*` and an `align-*`**, because items sit in two-dimensional cells:

- **inline axis** (row direction, left→right in English) → `justify-*`
- **block axis** (column direction, top→bottom) → `align-*`

| Property | On | Aligns | Default |
|---|---|---|---|
| `justify-items` / `align-items` | container | every item **inside its cell** | `stretch` (unless the item has a set size or an aspect ratio — a fixed width-to-height shape — e.g. images → `start`) |
| `justify-self` / `align-self` | item | one item inside its cell | inherits from `*-items` |
| `justify-content` / `align-content` | container | the **whole grid of tracks** inside the container, when the tracks are smaller than it | `normal` (tracks packed at the start; `auto` tracks stretch to fill) |
| `place-items`, `place-self`, `place-content` | — | shorthands: `align` value first, then `justify` | — |

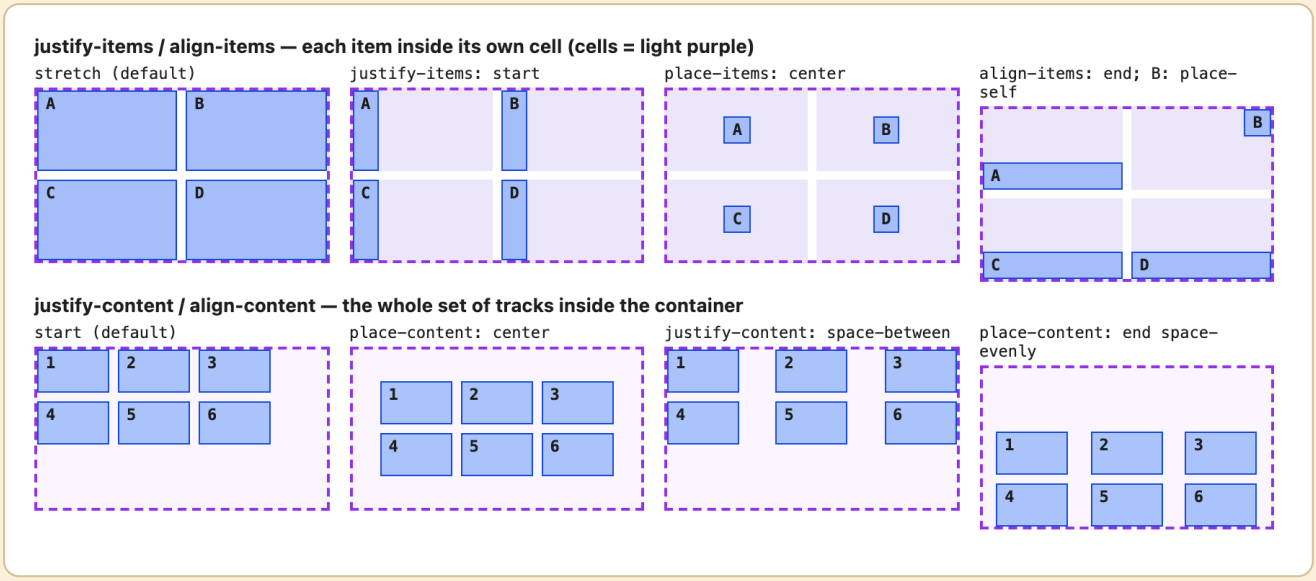



*Figure 9.5 — `code/05_alignment.html`. The light-purple rectangles are the cells (painted with background gradients), so you can see each item's position inside its cell.*

Key points:

- `place-items: center` on a grid is the shortest centring recipe in CSS: `display: grid; place-items: center`.
- **`*-content` only does something when there is free space** — tracks with fixed sizes smaller than the container. With `fr` tracks there's no free space and it has no effect.
- `space-between` / `space-around` / `space-evenly` in `*-content` **widen the effective gaps**; items spanning several tracks get bigger too.
- Auto margins work in grid as well: `margin-left: auto` pushes an item to the end of its cell.

### Self-check

1. Centre one item inside its cell without affecting others. → `place-self: center` on that item.
2. Why does `justify-content: center` do nothing on `grid-template-columns: 1fr 1fr`? → `fr` tracks use up all the space; nothing is left to distribute.
3. What order do values in `place-items: end start` follow? → `align-items: end`, `justify-items: start`.

## 9.7 Subgrid

### The problem it solves

A grid item that is itself a grid has its *own* independent tracks. Three pricing cards side by side each lay out title, text and button, but card 2's longer title pushes its text lower — the rows don't line up across cards. Before subgrid, fixes involved fixed heights or JavaScript.

### What it is

`grid-template-rows: subgrid` (or `grid-template-columns: subgrid`) on a grid item that spans several tracks of its parent tells it: **don't create your own tracks on this axis — reuse the parent's tracks I'm sitting on**. The children of the card then take part in the parent's row sizing, so the tallest title in *any* card sets the title row height for *all* cards. Subgrid is defined in CSS Grid Level 2 (the second version of the Grid specification) and has been **Baseline since September 2023** (Firefox 71, Safari 16, Chrome/Edge 117); by 2026 it is "widely available".

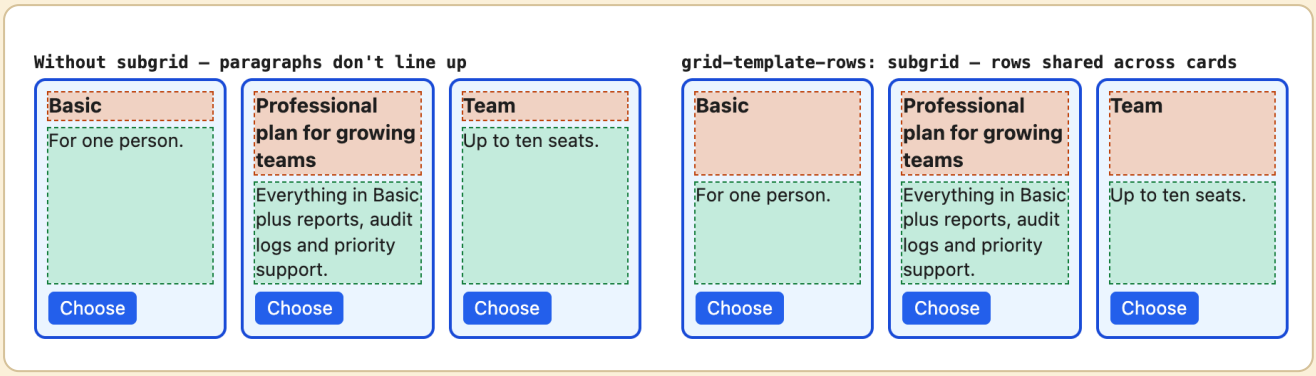



*Figure 9.6 — `code/06_subgrid.html`. Orange dashed = title, green dashed = paragraph.*

```
.sub       { display: grid; grid-template-columns: repeat(3, 1fr);
             grid-template-rows: auto auto auto; row-gap: 10px; }  /* the parent owns 3 rows */
.sub .card { display: grid; grid-row: span 3;                       /* each card covers all 3 */
             grid-template-rows: subgrid; row-gap: 6px; }           /* ...and reuses them */
```

Measured by `code/06_subgrid.mjs` (top edge of each paragraph and link):

```
plain  <p> tops: 76.9, 114.7, 76.9    link tops: 190.9, 190.9, 190.9
sub    <p> tops: 114.7, 114.7, 114.7  link tops: 190.9, 190.9, 190.9
```

Without subgrid the links happen to align (each card's middle row is `1fr` and the cards stretch to equal height), but the paragraphs don't. With subgrid every row is shared.

### Details

- The subgrid item must **span** the parent tracks it wants to share (`grid-row: span 3`).
- A subgrid inherits the parent's gaps but may override them (`row-gap: 6px` above).
- You can subgrid one axis and define your own tracks on the other.
- Line names from the parent are available inside the subgrid.
- Typical uses: card rows, forms where labels and inputs in separate `<fieldset>`s (HTML groups of related form controls, Chapter 5) must align, a list whose items have several aligned columns (a `<ul>` of rows with `grid-template-columns: subgrid`).

### Self-check

1. What must a subgrid item do besides `grid-template-rows: subgrid`? → Span the parent's rows it wants to use.
2. Can a subgrid have a different gap? → Yes, it can override the inherited gap.

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## 9.8 Grid vs Flexbox

| Question | Flexbox | Grid |
|---|---|---|
| Dimensions | One — a row **or** a column | Two — rows **and** columns |
| Who decides sizes? | **Content-out**: items' content and flex factors | **Layout-in**: the container's track definitions |
| Items in different rows line up in columns? | No — each flex line is independent | Yes — that's the point |
| Wrapping | Each line sizes itself; a lonely last item stretches | Items drop into the same columns; last row keeps the column width |
| Overlapping items | Needs negative margins or positioning | Place two items in the same cell |
| Gaps | `gap` | `gap` |
| Reordering | `order`, `*-reverse` | Line/area placement, `order`, `dense` |
| Typical jobs | Navbars, toolbars, button groups, tags, media objects, centring one thing, "push the last item right" | Page layouts, galleries, card grids, dashboards, form layouts, anything designed on a grid |
| Aligning content across siblings | Not possible | `subgrid` |

Rules of thumb:

1. **Is the layout driven by the content (a row of things of different sizes)?** → Flexbox.
2. **Do things need to line up in both directions, or do you know the track sizes up front?** → Grid.
3. **Use both**: grid for the page and the card gallery, flexbox inside each card's footer to space its buttons. They nest freely.

### In the interview

**30-second answer:** "Flexbox is one-dimensional and content-out: items sit in a row or column and share space based on their content and flex factors. Grid is two-dimensional and layout-in: I define rows and columns and place items into them, so things align across rows. I use grid for page structure and card grids — for example `repeat(auto-fill, minmax(200px, 1fr))` for a responsive gallery — and flexbox for components like navbars and button groups. They're often nested."

- *Can you make a responsive grid without media queries?* → Yes: `repeat(auto-fit | auto-fill, minmax(min(200px, 100%), 1fr))`.
- *Difference between auto-fill and auto-fit?* → Only visible when items don't fill a row: auto-fill keeps empty tracks, auto-fit collapses them so items stretch.
- *Why are my `1fr` columns unequal?* → `1fr` has an `auto` minimum (content size); use `minmax(0, 1fr)`.

---

## 9.9 Patterns

### Holy grail layout (header, nav, main, aside, footer)

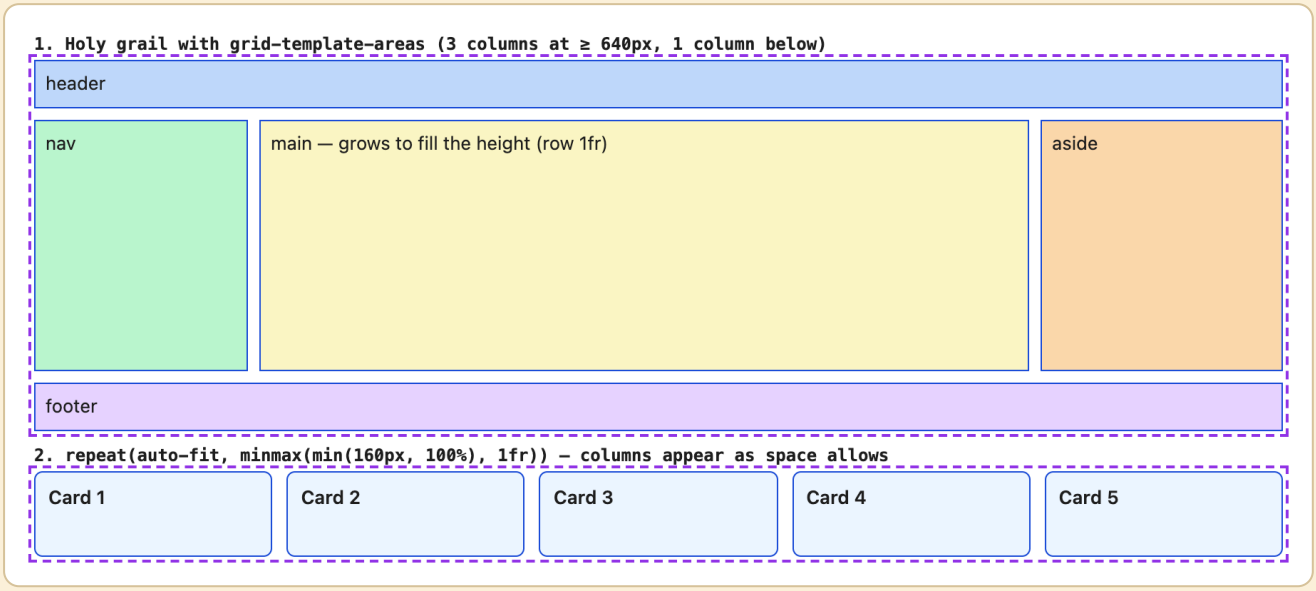
```
/* mobile first: one column, in source order */
.page { display: grid; gap: 8px; min-height: 100dvh;          /* demo uses 260px */
        grid-template-areas: "header" "nav" "main" "aside" "footer";
        grid-template-rows: auto auto 1fr auto auto; }
.page > header { grid-area: header; }  .page > nav    { grid-area: nav; }
.page > main   { grid-area: main; }    .page > aside  { grid-area: aside; }
.page > footer { grid-area: footer; }

/* three columns when there's room */
@media (min-width: 640px) {
  .page { grid-template-columns: 150px 1fr 170px;
          grid-template-rows: auto 1fr auto;
          grid-template-areas:
            "header header header"
            "nav    main   aside"
            "footer footer footer"; }
}
```

The `1fr` row makes `main` take all spare height, so the footer sits at the bottom of the screen (the sticky footer of §8.7, done with grid). Media queries are covered in Chapter 10.

### Responsive card grid (RAM)

```
.cards { display: grid; gap: 10px;
         grid-template-columns: repeat(auto-fit, minmax(min(160px, 100%), 1fr)); }
```



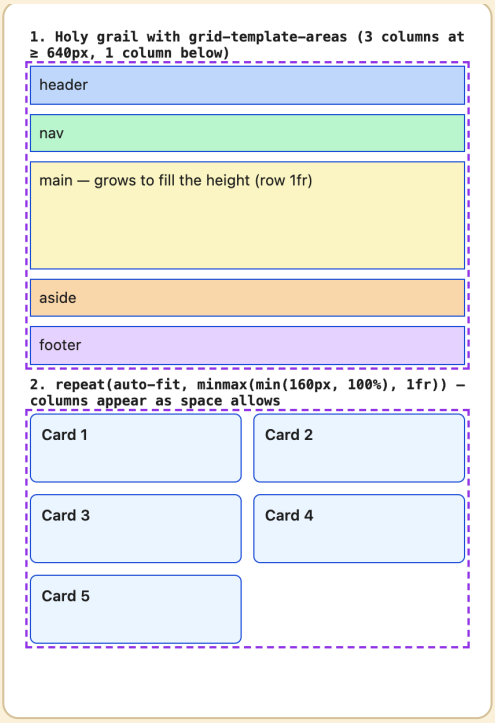



*Figure 9.7 — `code/07_patterns.html` screenshotted at 900px and 400px wide.*

`code/07_patterns.mjs` loads the page at both widths:

```
viewport 900px
  holy grail: 150px 540px 170px
  cards     : 167.188px 167.203px 167.203px 167.203px 167.188px
viewport 400px
  holy grail: 376px
  cards     : 183px 183px
```

At 900px the container is 876px: 876 − 150 − 170 − 2 × 8 = 540 for `main`; the cards fit floor(886 / 170) = 5 columns. At 400px the media query no longer matches (one 376px column) and only 2 card columns fit. Card 5 keeps the column width on its own row — the thing flex-wrap couldn't do (§8.6).

### Other patterns worth knowing

| Pattern | Grid recipe |
|---|---|
| Centre anything | `display: grid; place-items: center` |
| Overlay / hero caption | Image and caption both `grid-area: 1 / 1`; align the caption with `align-self` / `justify-self` |
| Full-bleed article | Named lines `[full-start] 1fr [content-start] minmax(0, 60rem) [content-end] 1fr [full-end]` (9.4) |
| Form with aligned labels | `grid-template-columns: max-content 1fr` on the form, one label + input per row |
| Sidebar that collapses | `grid-template-columns: fit-content(250px) minmax(0, 1fr)` |
| 12-column design system grid (the column grid a team's shared design rules are built on) | `repeat(12, minmax(0, 1fr))` + `grid-column: span 4` on items |

### Masonry: not yet

Pinterest-style "masonry" (items packed into columns of uneven height) is being standardised in CSS Grid Level 3 under the name **`display: grid-lanes`**. As of 2026 it has shipped in Safari (26.4) and is behind flags (hidden experimental settings users must switch on) in Chrome and Firefox — not Baseline. Use CSS multi-column (`columns: 3`, newspaper-style columns) or a JavaScript library, and wrap any

experiment in `@supports (display: grid-lanes)` (a feature query: the rules inside apply only if the browser supports that feature).

---

## 9.10 Debugging, pitfalls and interview prep

### Debugging

All major browsers show a **grid badge** beside grid containers in the Elements/Inspector panel of DevTools (the built-in developer tools, F12). Toggling it overlays line numbers (positive and negative), track sizes, gaps and area names — the same information our red badges draw. Firefox's grid inspector pioneered this and is still the most detailed.

### Pitfalls at a glance

| Symptom | Cause | Fix |
|---|---|---|
| `1fr` columns are unequal / grid overflows on mobile | `1fr` = `minmax(auto, 1fr)`; long words, `<pre>`, images or tables set the minimum | `minmax(0, 1fr)`; `min-width: 0` or `overflow` on the item |
| Horizontal scrollbar with `%` columns | `%` tracks ignore gaps | Use `fr` |
| Single auto-fill column overflows narrow screens | Minimum larger than the container | `minmax(min(200px, 100%), 1fr)` |
| Item lands in an unexpected extra row/column | Placed beyond the explicit grid → implicit tracks; or a typo in an area name (unknown names create implicit lines) | Check line numbers in DevTools; spell area names consistently |
| `grid-template-areas` ignored | Strings have different cell counts or a non-rectangular area — the whole declaration is invalid | Fix the ASCII art |
| An image doesn't fill its cell | Items with an aspect ratio (images, video) treat the default alignment as `start`, not `stretch` — checked in Chrome: a 100×50 image in a 200×300 cell stays 100×50 | `width: 100%; height: 100%; object-fit: cover` (scale the image to fill the box, cropping the overflow; Chapter 11) |
| `justify-content` does nothing | `fr` tracks leave no free space | Use fixed tracks, or align items instead |
| Tab order jumps around | `dense`, placement or `order` moved items visually | Keep DOM order = visual order |

### In the interview

**30-second answer:** "Grid is two-dimensional layout. On the container I define tracks with `grid-template-columns` and `-rows` using lengths, `fr` for fractions of leftover space, `minmax`, and `repeat` — `repeat(auto-fill, minmax(200px, 1fr))` gives a responsive grid with no media queries. Items are auto-placed into cells, or placed by line numbers, spans or named `grid-template-areas`. Extra items create implicit tracks sized by `grid-auto-rows`. Alignment uses `justify-*` for the inline axis and `align-*` for the block axis, and `subgrid` lets nested grids share the parent's tracks so cards line up."

Follow-ups:

- *How is `fr` calculated?* → Container size minus fixed tracks and gaps, divided by the total of the `fr` values; tracks whose share is below their minimum are fixed and the rest re-split. 700 − 200 −

40 = 460 → 1fr = 153.33.

- *auto-fill vs auto-fit?* → Same until there are too few items; then auto-fill keeps empty tracks, auto-fit collapses them.
- *How would you build a layout that reorders at a breakpoint?* → Named areas, rewritten in a media query — while keeping DOM order sensible for keyboard and screen-reader users.
- *What is subgrid for?* → Aligning the insides of sibling components (card titles, form labels) to shared tracks.

### Self-check

1. `grid-template-columns: repeat(3, 1fr)` overflows on mobile because of a wide table. Two fixes? → `repeat(3, minmax(0, 1fr))` and let the table scroll (`overflow-x: auto` on a wrapper); or fewer columns at small widths.
2. Which property sizes rows that you did not declare? → `grid-auto-rows`.
3. Shortest way to centre a child in both axes? → `display: grid; place-items: center` on the parent.

---

## Chapter 9 TL;DR

- **Grid** = two-dimensional, layout-in. `display: grid` → direct children are items. Vocabulary: lines (1…n+1, and −1 backwards), tracks, cells, areas, gaps; explicit vs implicit grid.
- **Tracks**: px/%/`fr`/`auto`/`min-content`/`max-content`/`fit-content()`/`minmax()`. **`fr` = share of space left after fixed tracks and gaps** (700 − 200 − 40 = 460 → 153.33 / 306.67, verified). `1fr` = `minmax(auto, 1fr)` → use `minmax(0, 1fr)` for truly equal columns.
- **RAM**: `repeat(auto-fill|auto-fit, minmax(min(X, 100%), 1fr))`; n = floor((W + gap) / (X + gap)). auto-fill keeps empty tracks, auto-fit collapses them.
- **Placement**: `grid-column: 1 / 3`, `span 2`, `1 / -1` (full width), `grid-area: row-start / col-start / row-end / col-end`, named lines, and `grid-template-areas` ASCII art (rectangles only, `.` = empty). Items can overlap in one cell.
- **Implicit grid**: `grid-auto-rows/columns` size extra tracks; `grid-auto-flow: row | column | dense` (dense breaks visual/DOM order).
- **Alignment**: `justify-*` = inline axis, `align-*` = block axis; `*-items`/`*-self` inside cells, `*-content` for the whole grid when tracks leave free space; `place-*` shorthands; `place-items: center` centres.
- **Subgrid** (Baseline since Sept 2023): `grid-template-rows: subgrid` on a spanning item reuses the parent's tracks so card internals align.
- **Grid vs Flexbox**: grid for page structure and uniform grids; flexbox for content-sized rows inside components; nest them.
- **Masonry** (`display: grid-lanes`) is Safari-only in 2026 — not Baseline.

Next: Chapter 10 — Responsive Design.